CapExIQ

An AI-Enabled Capital-Budgeting and Investment Decision Platform

Field	Detail
Project topic	Topic 9 — AI Capital-Budgeting Dashboard
Flagship case	NovaRetail GCC — Automated Micro-Fulfilment Centre, Dubai (installing automation technology)
Decision evaluated	AED 24,000,000 capital commitment over a six-year horizon at an 11.50% WACC
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Date	5th August 2026
Word count	1,645 words (excluding tables, headings and front matter)

1. Executive Summary

NovaRetail GCC, a hypothetical UAE omnichannel retailer, is evaluating an AED 24.0 million investment in an Automated Micro-Fulfilment Centre in Dubai: AED 22.0M of capital expenditure and AED 2.0M of working capital, over a six-year economic life. CapExIQ is the decision-support platform built to evaluate it. All financial mathematics is executed by a deterministic TypeScript engine; artificial intelligence is confined to explanation, challenge and drafting, and never computes a number that reaches a decision.

The base case returns a net present value of AED 12,083,628, an internal rate of return of 26.30% against an 11.50% hurdle, a modified IRR of 19.34%, a profitability index of 1.5035 and payback in 3.10 years (3.98 discounted). Every decision rule is satisfied and the engine returns Approve. The recommendation is to approve, subject to two conditions drawn from the sensitivity analysis rather than from convention.

CapExIQ is not a single-purpose calculator. It ships eight investment archetypes spanning the capital decisions named in the brief, and sixteen AI capabilities, each degrading to a deterministic fallback when no model is available.

Metric	Result	Benchmark	Margin	Verdict
Net present value @ 11.5%	AED 12,083,628	> 0	—	Creates value
Internal rate of return	26.30%	11.50%	+14.80 pp	Pass
Modified IRR	19.34%	11.50%	+7.84 pp	Pass
Profitability index	1.5035x	1.00x	+0.50x	Pass
Payback	3.10 years	4.00 years	0.90 yr	Pass
Discounted payback	3.98 years	4.50 years	0.52 yr	Pass

2. The Financial Problem

Question 1 — What financial decision is being analysed, and why does it matter?

NovaRetail fulfils online orders by manual picking from store back-rooms. Direct fulfilment costs AED 14.50 per order, preparation is slow, and picking capacity now constrains growth in precisely the urban Dubai postcodes where demand is expanding fastest. The proposed centre replaces manual picking with goods-to-person robotics and automated storage and retrieval, reducing direct cost to AED 4.20 per order and lifting capacity to 8,000 orders per day.

Management must choose between four mutually exclusive options: commit the full AED 24.0M now; release AED 14.0M and gate the remaining AED 10.0M on measured Year-1 savings; defer twelve months pending vendor performance evidence; or reject and retain manual fulfilment. The decision is material because AED 24.0M is a significant share of the annual capital budget, the assets are highly specific and difficult to redeploy, and the error is asymmetric in both directions: approving a weak case strands capital in illiquid robotics, while rejecting a strong one forfeits a defensible position in a market that is consolidating quickly.

3. Corporate Finance Concepts and Formulas

Question 2, part one — which concepts and formulas are required?

The evaluation discounts incremental after-tax free cash flows at the weighted average cost of capital. Five formulas carry the decision.

Concept	Formula	Decision rule
Free cash flow	FCFt = (EBITDAt − Dt)(1 − τ) + Dt − ΔNWCt − CapExt + Terminalt	Depreciation is deducted to find taxable income, then added back as a non-cash charge — the tax shield
Net present value	NPV = Σ FCFt ÷ (1 + r)^t, for t = 0 to N	Accept above zero. Primary rule
Internal rate of return	The rate r* where NPV(r*) = 0	Accept above the hurdle rate
Modified IRR	MIRR = (FV inflows at reinvestment rate ÷ PV outflows at finance rate)^(1/N) − 1	Corrects IRR’s assumption that interim flows earn the project’s own return
Profitability index	PI = PV of inflows ÷ initial outlay	Accept above 1.0. Ranks projects under capital rationing
Payback / discounted	Years until cumulative (discounted) cash flow turns positive	Liquidity screen only — measures exposure, not value
Cost of capital	r = (E/V)·re + (D/V)·rd(1 − τ)	Cost of equity built from CAPM plus explicit risk premiums

These concepts are archetype-independent, which is the analytical basis for generalising the platform: the mathematics does not care what generated the cash flows, only that they are incremental, after-tax and correctly timed.

4. Data and Assumptions

Question 2, part two — what data and assumptions are required?

Every input is classified by provenance, so a reader can see immediately which numbers are observed, which are forecast and which are judgement.

Classification	Parameters	Source
Historical	Shipping performance, delivery lateness, order profitability	DataCo Smart Supply Chain (Mendeley 8gx2fvg2k6, CC BY 4.0)
Current external	UAE corporate tax 9% above AED 375,000; DEWA tariffs 23/28/32/38 fils per kWh plus 6 fils surcharge and 5% VAT; three-month EIBOR 3.79%	Federal Decree-Law 47/2022; DEWA 2026 published tariffs; Central Bank of the UAE
Forecast	Year-1 savings AED 7.5M (+4% p.a.); contribution margin AED 2.5M (+5%); OpEx AED 2.2M (+3%); salvage AED 2.0M	Management forecast, corroborated by the platform’s capacity model
User-entered	CapEx AED 22.0M (equipment 18.0, installation 2.5, software 1.2, training 0.3); working capital AED 2.0M; life 6 years; WACC 11.50%	Editable model inputs

Classification	Parameters	Source
AI-generated	Narrative explanation, risk framing, scenario proposals, memorandum drafts	Server-side routes — advisory only, never numeric

The hurdle rate is derived, not asserted. Cost of equity = 4.20% risk-free + (1.15 × 6.00% equity risk premium) + 0.75% UAE country risk premium + 3.50% project execution premium = 15.35%. Cost of debt = 3.79% EIBOR + 2.50% credit spread = 6.29% pre-tax, 5.72% after tax at 9%. At a 60/40 equity-debt structure: (0.60 × 15.35%) + (0.40 × 5.72%) = 11.50%. The execution premium is the honest component — this is a first-of-kind robotics deployment, and pricing it at the corporate average would understate the risk.

The critical assumption is the AED 7.5M of Year-1 savings, which drives roughly three-quarters of the value. It is not asserted. The platform derives it from a labour bridge: 75.2 manual full-time equivalents against 8.3 automated, giving 66.9 displaced at an all-in loaded cost of AED 112,000, or AED 7,491,556 — within 0.11% of the forecast. NovaRetail GCC is explicitly hypothetical; the DataCo data is genuine external operational data and is never presented as its financial statements.

5. Financial Calculations

Depreciation is straight-line to salvage: (AED 22.0M − 2.0M) ÷ 6 = AED 3,333,333 per year, shielding AED 300,000 of tax annually. Year-1 EBITDA is AED 10.0M of benefits less AED 2.2M of operating cost = AED 7.80M; EBIT AED 4,466,667; tax at 9% AED 402,000; operating cash flow AED 7,398,000.

AED	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Total benefits	10,000,000	10,425,000	10,868,250	11,330,543	11,812,705	12,315,601
Less operating cost	(2,200,000)	(2,266,000)	(2,333,980)	(2,403,999)	(2,476,119)	(2,550,403)
EBITDA	7,800,000	8,159,000	8,534,270	8,926,543	9,336,585	9,765,198
Free cash flow	**7,398,000**	**7,724,690**	**8,066,186**	**8,423,154**	**8,796,293**	**13,186,330**
Discount factor	0.8969	0.8044	0.7214	0.6470	0.5803	0.5204
Present value	6,634,978	6,213,429	5,818,936	5,449,734	5,104,172	6,862,380

Year 0 carries the AED 24,000,000 outflow; Year 6 includes AED 2.0M salvage and AED 2.0M of recovered working capital. Present value of inflows totals AED 36,083,628 against a AED 24,000,000 outlay, giving NPV of AED 12,083,628 and PI of 1.5035. IRR solves to 26.30% by Newton-Raphson with a bisection fallback. MIRR of 19.34% sits below IRR precisely because it reinvests interim flows at 11.50% rather than at the project's own return, which is the more defensible assumption. Cumulative cash flow turns positive during Year 4. All five rules agree, and the outcome is pinned by an automated golden-value regression test so no published figure can drift from the engine.

6. Artificial Intelligence Features

Question 4 — how does AI improve the analysis, dashboard, forecasting and decision-making?

AI improves the work in four ways: it compresses drafting from hours to seconds, it interrogates assumptions from perspectives an author would not volunteer, it makes the model addressable in natural language by directors who will not open a spreadsheet, and it structures unstructured inputs such as vendor quotations. It does not improve the arithmetic and is deliberately prevented from touching it. Sixteen capabilities are implemented; each degrades to a deterministic fallback, so the platform is fully functional with no API key.

Capability	What it does	Information used	Result produced	How it helps	Limitation / risk
Finance assistant	Answers free-text questions about the live model	Current assumptions and computed metrics as injected context	Plain-language explanation	A non-financial director can interrogate the model directly	May misread ambiguous questions; not auditable line by line
Result explanation	Explains why a metric moved	Before-and-after metric pair	Short causal narrative	Removes the analyst from routine explanation	Correlation can be narrated as causation
Board recommendation	Drafts a structured decision with drivers, risks and controls	Validated metrics, scenarios and risk alerts	Schema-constrained JSON with a confidence rating	Compresses memo preparation from hours to seconds	Will write a confident memo from poor inputs; sign-off is mandatory
Threat radar	Ranks risk axes specific to the investment type	Archetype profile plus computed metrics	Severity-scored axes with mitigations and the driver each maps to	Surfaces archetype-specific exposure a generic checklist misses	Only detects risks encoded in its priors
Board debate	Simulates a capital committee — CFO, COO, Risk Officer, sceptical non-executive	Metrics, scenario set, archetype persona slants	Four opposing arguments and a synthesis	Manufactures the challenge a single author cannot supply	Simulated dissent is not real governance
Board memorandum	Drafts the full memorandum in structured sections	Metrics, scenarios, risk register, archetype KPI set	Sectioned board document	Produces consistent, submission-ready language	Fluent prose lends false authority to weak assumptions
Scenario studio	Proposes additional named scenarios beyond the standard three	Archetype scenario themes and base-case posture	Assumption multiplier sets for the engine to evaluate	Widens the stress space past the conventional trio	Proposes assumptions only; it never produces results
ESG impact	Green-financing and sustainability commentary	Capex composition and archetype relevance	ESG narrative, or an explicit not-applicable flag	Prevents fabricated ESG content on archetypes where it is meaningless	Not an assurance opinion and no substitute for one
Quote parser	Extracts structured capex lines from pasted vendor quotations	Raw quotation text, archetype capex categories	Line items with amounts, categories and a confidence score	Replaces manual re-keying and its transcription errors	Ambiguous lines are returned unparsed rather than guessed
Retrieval-grounded answers	Retrieves the passages of project documentation that bear on a question before answering	A 90-passage corpus built from the methodology, assumption register and scenario definitions	A streamed answer in which every claim carries a numbered citation to its source extract	A reader can verify each statement against the document it came from rather than trusting it	Retrieval can miss a relevant passage; a citation proves provenance, not correctness
Macro reference panel	Presents the external rate, tariff and tax basis behind the model	Values transcribed by hand from published sources, each carrying its citation and as-of date	A reference panel explicitly flagged as not live	External inputs are auditable to a named public document	Values are point-in-time and must be re-transcribed as sources are revised
Driver attribution	Decomposes a change in net present value into the contribution of each assumption	Two assumption sets and the engine output for each	A ranked contribution per driver that reconciles exactly to the total movement	Answers why a number moved, not merely that it moved	Attributed shares depend on a fixed substitution order, which is documented rather than chosen per call
Fitted cash-flow forecast	Fits the benefit growth rate to an observed series and projects it with uncertainty bands	Historical operational spend, by least squares on log values	P10, P50 and P90 paths whose width derives from the regression standard error	Replaces an assumed growth rate with a fitted one, and shows how uncertain it is	A log-linear trend fit with no seasonality or autocorrelation term; it declines to fit fewer than four observations
Assumption screening	Screens inputs against plausible ranges before the engine evaluates them	The assumption register and its stated ranges	Severity-ranked findings naming the expectation breached and the consequence	Catches an implausible input before it is discounted, charted and written into a memorandum	Range checks detect only what they were written to detect
Governing-measure selection	States which decision rule governs the comparison at hand and which would mislead	Horizon equality, capital-rationing status and the cash-flow sign pattern	A primary measure, supporting measures, and cautions with reasons	Prevents the common error of applying correct arithmetic to the wrong comparison	Advisory on rule choice only; it does not evaluate the alternatives
Voice intent	Maps a spoken command to a structured application action	Transcribed text plus workspace context	Typed intent with parameters, or an explicit unknown	Enables hands-free navigation during a board session	Returns unknown rather than guessing at low confidence

The governance boundary is architectural, not advisory. Every route receives pre-computed figures and is instructed never to recalculate; every response is labelled as advisory and carries its provenance, so a reader can see whether they are looking at a model response or the deterministic fallback. User text is screened for instruction-override attempts and redacted for personal data before it is forwarded, and outbound network access is restricted to the model provider alone, so the platform cannot retrieve third-party web content even if asked to.

Assistant question and answer transcript

Question put to the assistant	Answer returned, with its basis
Why did the NPV decrease?	Because the scenario changed. Moving from Base to Pessimistic applies a 0.75x multiplier to operating benefits, 1.15x to capital and operating cost, and lifts the discount rate to 14.50%. Net present value moves from AED 12,083,628 to negative AED 4,940,625, a swing of AED 17.02M, of which the benefit multiplier is by far the largest single term.
What happens if the discount rate rises?	Value falls, and the break point is the internal rate of return. Net present value reaches exactly zero at 26.30%. Below that rate the project creates value; above it the project destroys value. The base case therefore carries 14.80 percentage points of headroom on the cost of capital.
Which assumption has the greatest effect on the result?	Total operating benefits, by a factor of two over anything else. Flexed by an identical plus or minus 20% so the comparison is fair, benefits swing net present value by AED 16.67M — more than the entire base-case value. Project life follows at AED 8.39M, capital expenditure AED 8.25M, cost of capital AED 5.17M, operating cost AED 3.57M and salvage AED 0.37M.
Should management accept or reject the project?	Accept, with gates. All five decision rules agree in the base case and expected value across probability-weighted scenarios is AED 9,560,152. Two conditions attach: release capital against measured Year-1 savings rather than against the calendar, and require a vendor performance guarantee before commitment. The engine returns Approve; the decision remains the committee's.
Explain the result to a non-financial manager.	The company spends AED 24.0M today. Over six years the savings and margin it generates are worth AED 36.1M in today's money, so the company ends up about AED 12.1M better off than if it did nothing. The money spent is recovered in a little over three years. The main thing that could go wrong is the savings arriving smaller than promised — if they come in more than 29% below forecast, the project stops being worth doing.

7. The Dashboard and Platform Architecture

CapExIQ is a Next.js application with a strictly-typed deterministic finance engine at its core. The executive dashboard carries the six required components: key-performance cards for NPV, IRR, MIRR, PI and payback; an annual free-cash-flow chart with a cumulative line crossing zero in Year 4; a three-scenario comparison; a sensitivity module with a normalised tornado chart and two-dimensional NPV heatmaps; a ranked risk-alert panel; and the AI recommendation panel. Supporting routes cover the assumptions register, a 5,000-iteration Monte Carlo simulation, a portfolio optimiser solving the 0-1 knapsack under capital rationing, covenant analysis and a print-ready board memorandum.

The architecture separates cash-flow construction from evaluation. An archetype configuration produces an annual benefit profile; the audited engine then evaluates it. Because that boundary is clean, the same tested mathematics serves eight distinct investment types without being forked, and the flagship case reproduces its figures exactly after the refactor.

Investment archetype	Outlay	NPV	IRR	Verdict
Installing automation technology (flagship)	AED 24.0M	+AED 12.08M	26.3%	Approve
Expanding a production facility	AED 26.7M	+AED 7.28M	16.4%	Phased
Purchasing new machinery	AED 5.21M	+AED 1.57M	18.9%	Approve
Building an AI platform	AED 7.80M	(AED 0.01M)	16.0%	Delay
Opening a new branch	AED 6.80M	(AED 0.10M)	12.1%	Delay
Introducing a new product	AED 6.30M	(AED 0.55M)	10.4%	Delay
Launching an online service	AED 6.90M	(AED 1.49M)	14.0%	Reject

Investment archetype	Outlay	NPV	IRR	Verdict
Entering a new market	AED 10.9M	(AED 2.21M)	10.7%	Reject

That five of the eight default cases fail is deliberate. A template pack in which every archetype is comfortably profitable would demonstrate optimistic defaults rather than a working engine.

Access is governed rather than cosmetic. Six executive roles map to twenty permissions, and authorisation is decided from a signed session cookie before a page renders, so the funding structure and the approval workflow cannot be reached by changing a setting in the interface.

8. Scenario and Sensitivity Analysis

Question 3 — what do the calculations show across the three scenarios?

	Optimistic	Base case	Pessimistic
CapEx / benefits / OpEx	×0.95 / ×1.10 / ×0.95	×1.00	×1.15 / ×0.75 / ×1.15
Discount rate	10.50%	11.50%	14.50%
Net present value	AED 19,013,977	AED 12,083,628	(AED 4,940,625)
Internal rate of return	33.59%	26.30%	8.23%
Profitability index	1.830x	1.504x	0.819x
Payback	2.63 years	3.10 years	5.06 years
Decision	**Approve**	**Approve**	**Reject**

Weighting the three outcomes at 50/25/25 gives an expected net present value of AED 9,560,152. The pessimistic case is not a mild downside: it destroys AED 4.94M and the engine returns a hard Reject, because it stacks a 25% benefit shortfall, a 15% cost overrun and a 300 basis-point rate rise simultaneously.

Flexing every driver by an identical ±20%, so the bars are genuinely comparable, total operating benefits dominates, swinging NPV by AED 16.67M — more than the project's entire base-case value. Project life follows at AED 8.39M, capital expenditure at AED 8.25M, the discount rate at AED 5.17M and operating cost at AED 3.57M. Salvage value is close to irrelevant at AED 0.37M, contributing 8.61% of NPV in present-value terms.

The decision changes when Year-1 benefits fall more than 29.0% below forecast, when total outlay exceeds AED 36.08M (a 50.4% overrun), or when the discount rate rises above 26.30%. The principal financial risk is therefore benefit realisation, not capital cost — and the correct management response is to condition capital release on measured savings rather than to negotiate harder on equipment price.

9. AI Limitations and Ethical Risks

Risk	Position taken
Accuracy of AI estimates	AI output is not verifiable the way a formula is. AI is therefore excluded from every calculation; the engine is reproducible TypeScript pinned by regression tests
Use of incorrect data	A confident narrative built on a wrong input is more dangerous than no narrative, because fluency reads as authority. Provenance classification makes every input auditable
Hallucination	Language models invent plausible figures. Mitigated by passing only validated output into prompts, instructing the model never to compute, and delimiting user text against injection
Confidentiality	No personal or customer data is transmitted. Only aggregated financial outputs leave the server and credentials remain server-side in route handlers

Risk	Position taken
Bias in forecasts	Forecasts inherit the optimism of whoever set them, and a model trained on published business cases amplifies it. The pessimistic scenario, the risk panel and the simulated board debate are deliberate counterweights
Human review	Every AI response is labelled advisory, carries a provenance badge, and requires sign-off before it informs a decision
Presenting generated content as observed	A generated figure displayed as a market observation is indistinguishable from a real one. Any panel that cannot cite a source is now labelled as not live and carries the as-of date of the document its values were transcribed from
Responsibility	Final responsibility for this investment decision rests with the Chief Financial Officer and the Capital Expenditure Committee. It does not rest with the AI system, and no output should be read as transferring it

One of these risks was found in our own build, which is the reason the row above exists. An earlier version of the macroeconomic panel prompted the model to act as a real-time data ingestion agent and published the result as observed market data. No external source was ever contacted: every interbank rate, utility tariff and lease price on that panel had been generated by the model and shown to a capital committee as fact. The generation step was removed and replaced with hand-transcribed values carrying citations and as-of dates. It is recorded here because it illustrates the central ethical risk precisely — not that a model states something false, but that a well-designed interface can make a generated figure look exactly like a measured one.

10. Final Recommendation

Question 5 — what recommendation, and what financial and AI risks attach to it?

Approve the investment. It returns AED 12.08M of net present value, an IRR of 26.30% against an 11.50% hurdle, a profitability index of 1.50 and payback inside four years. All five decision rules agree, expected value across weighted scenarios remains AED 9.56M, and the downside turns destructive only under a simultaneous three-factor stress.

Approval should carry two conditions, both drawn from the analysis rather than from habit. First, because benefit realisation dominates every other variable by a factor of two, release capital against measured Year-1 savings rather than against the calendar. Second, because the pessimistic case genuinely destroys value, require a vendor performance guarantee and a secondary-market equipment buyback before committing. Neither condition weakens the recommendation; both protect it.

Two categories of risk attach. The financial risks are benefit shortfall beyond 29.0%, a total outlay overrun beyond 50.4%, and a cost of capital above 26.30% — each individually survivable, dangerous only in combination. The AI-related risks are those set out in Section 9, and they are bounded by architecture: the model never touches the arithmetic, every response is labelled and traceable to its source, and the platform produces identical financial results whether or not the model is available at all. The decision, and the accountability for it, remain human.